

An exponential waiting model

The exponential distribution models a continuous random variable, often a nonnegative waiting time, using a positive rate.

Why does a nonnegative waiting time alone not establish an exponential model?

Rate and time

Model: print requests arrive as a Poisson process at 4 requests per hour.

What variable measures the waiting time until the next request?

Rate notation

The rate m , called the decay parameter, has units reciprocal to the time units.

$$X \sim \text{Exp}(m)$$

If time is measured in hours, what units should m have?

Mean waiting time

The exponential mean is the reciprocal of the rate.

$$\mu = \frac{1}{m}$$

At a rate of 4 requests per hour, what is the mean wait in minutes?

Converting a mean to a rate

Model: an exponential wait has a mean of 5 minutes.

What is the rate per minute?

The density

An exponential density decreases as the waiting time increases.

$$f(x) = me^{-mx} \quad (x \geq 0)$$

For $m=0.2$ per minute, which interval has more probability: 0 to 1 or 5 to 6 minutes?

The number e

The exponential density uses the constant $e \approx 2.71828$. The natural logarithm $\ln$ reverses exponentiation with base e .

Which calculator operation evaluates e raised to -0.8 ?

Sketching the exponential density

Model: $f(x) = 0.2e^{(-0.2x)}$ for $x \geq 0$, with x measured in minutes.

How would you label the axes and sketch its starting height and declining shape?

The upper tail

The exponential survival probability gives the chance of exceeding a cutoff.

$$P(X > x) = e^{-mx}$$

For $m=0.2$ per minute, what is the probability of waiting longer than 8 minutes?

A wait within a deadline

Model: an exponential wait has mean 5 minutes.

What is the probability that the wait is at most 8 minutes?

Exponential CDF

The cumulative distribution function (CDF) gives area to the left of x for $x \geq 0$.

$$F(x) = P(X \leq x) = 1 - e^{-mx}$$

For $m = 0.2$ per minute, what probability lies to the left of 8 minutes?

Between two waits

Model: an exponential wait has mean 5 minutes.

What is the probability of waiting between 2 and 6 minutes?

Choosing the tail

Model: an exponential processing time has mean 10 seconds. A job is late if it exceeds 15 seconds.

Which area describes a late job?

Units in the exponent

Model: arrivals have rate 6 per hour. A cutoff is 5 minutes.

How must the rate and cutoff units agree before calculating the tail probability?

An exponential percentile

Model: an exponential wait has mean 5 minutes.

What cutoff has 90% of waits at or below it?

An inverse exponential calculation

For an exponential rate m and probability $0 < a < 1$, the a -quantile solves $F(k) = a$.

$$k = -\frac{\ln(1 - a)}{m}$$

With $m = 0.2$ per minute, what calculation gives the median?

Mean and median

Model: an exponential wait has mean 5 minutes.

Is the median wait equal to 5 minutes?

Exponential spread

The exponential standard deviation equals its mean.

$$\sigma = \mu = \frac{1}{m}$$

What are the mean and standard deviation when $m=0.25$ per minute?

Expected total lifetime

Model: five independent components are used one after another. Each lifetime is exponential with mean 8 years.

What total operating lifetime is expected?

The memoryless property

For an exponential wait, the remaining wait has the original waiting-time distribution; t and s are nonnegative.

$$P(X > t + s \mid X > t) = P(X > s)$$

How does the probability statement describe a wait already longer than t ?

A remaining wait

Model: an exponential wait has mean 5 minutes. It has already exceeded 7 minutes.

What is the probability of waiting more than 3 additional minutes?

Comparing waiting models

Model A has exponential waits with mean 6 minutes. Model B has uniform waits from 0 to 12 minutes.

After waiting 11 minutes, would the two models give the same remaining-wait distribution?

Counts or waiting time?

Model: requests arrive as a Poisson process at 6 per hour.

Would the number of requests in 20 minutes use a Poisson or exponential model?

A Poisson connection

For a Poisson process, the count in a fixed interval has a Poisson distribution. Synthetic rate: 6 requests per hour.

What is the mean number of requests in 20 minutes?

Computing the linked count probability

Model: independent times between requests are exponential with rate 6 per hour.

What is the probability of exactly two requests during 20 minutes?

An empty interval

Model: requests arrive as a Poisson process at 6 per hour.

How does zero requests in 20 minutes relate to waiting longer than 20 minutes?

Model assumptions

A homogeneous Poisson process has a constant arrival rate and independent counts in disjoint time intervals.

Would sharply higher arrival rates at lunchtime fit this model over an entire day?

Preparing for averages

Model: independent exponential processing times have mean 10 seconds.

Would one processing time and the average of 100 processing times have the same spread?